

Roll No.....

Mid Semester Examination

Name of the Program: **B.Tech (ECE)**

Semester: **V**

Name of the Course: **Communication Systems II**

Course Code: **TEC 502**

Time: **90 minutes**

Maximum Marks: **50**

Note:

- (i) Answer **all the questions** by choosing *any one of the sub questions*.
- (ii) Each question carries **10 marks**.

Q1	(10 marks)	
(a)	(i) Given the signal $m(t) = 10 \cos(2000\pi t) \times \cos(8000\pi t)$. What is the minimum sampling rate based on the bandpass sampling theorem. (ii) Find the Nyquist sampling rate and Nyquist interval for each of the following signals (i) $\sin c(100\pi t)$ (ii) $\sin c^2(100\pi t)$.	CO1
OR		
(b)	A signal is ideally sampled with a sampling period of 50 μ sec and the pass through an ideal LPF with cutoff frequency of 15 KHz. What frequency components will appear in the LPF output?	
Q2	(10 marks)	
(a)	(i) Three analog signals, having bandwidths 1200 Hz, 600 Hz and 600 Hz, are sampled at their respective Nyquist rates, encoded with 12 bit words, and time division multiplexed. Determine the bit rate for the multiplexed signal. (ii) In a binary PCM system, the output signal to quantization noise ratio is to be held to a minimum value of 40 dB. Determine the number of required levels, and find the corresponding output signal to quantization noise ratio.	CO1
OR		
(b)	Derive an expression for signal to quantization noise ratio for a PCM system which employs uniform quantization technique. Given that input signal to the PCM system is a sinusoidal signal.	
Q3	(10 marks)	
(a)	What is the μ law of companding? Give its expression and draw it.	CO1
OR		
(b)	Illustrate and describe the types of uniform quantizer?	
Q4	(10 marks)	
(a)	Explain Differential Pulse Code Modulation (DPCM) in detail with suitable diagram.	CO1 & CO2
OR		
(b)	What are the properties of line coding?	
Q5	(10 marks)	
(a)	Consider a binary sequence 110101. Draw the waveforms for the following signaling formats. (i) Unipolar NRZ signaling (ii) Polar NRZ signaling (iii) AMI signaling (iv) Bipolar NRZ signaling	CO1 & CO2
OR		
(b)	For a sinusoidal modulating signal $m(t) = A \cos 2\pi f_m t$, prove that the maximum output signal to quantization noise ratio in a DM system under the assumption of no slope overload is given by	

$(SNR)_o = \left(\frac{S}{N_q} \right)_0 = \frac{3 f_s^3}{8 \pi^2 f_m^2 f_M}, \text{ where } f_s = \frac{1}{T_S} \text{ is the sampling rate and } f_M$ is the cutoff frequency of a low pass filter at the output of the receiver.	
---	--